Key features of parabolas

1 Consider the general quadratic equation $y = ax^2 + bx + c$, $a \neq 0$.

- **a** If $a < 0$, in what direction will the parabola open?
- **b** If $a > 0$, in what direction will the parabola open?

2 Does the parabola represented by the equation $y = x^2 - 8x + 9$ open upward or downward?

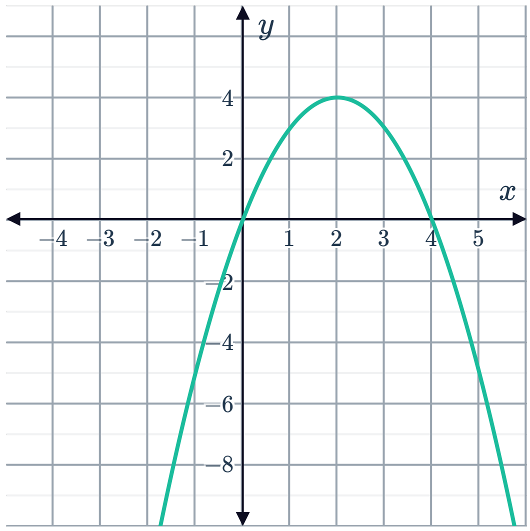
3 Does the graph of $y = x^2 + 6$ have any x-intercepts? Explain your answer.

4 State whether the following parabolas have x-intercepts:

- **a** $y = (x-7)^2 + 4$
- **b** $y = -(x-7)^2 + 4$
- **c** $y = -(x-7)^2 - 4$
- **d** $y = (x-7)^2 - 4$

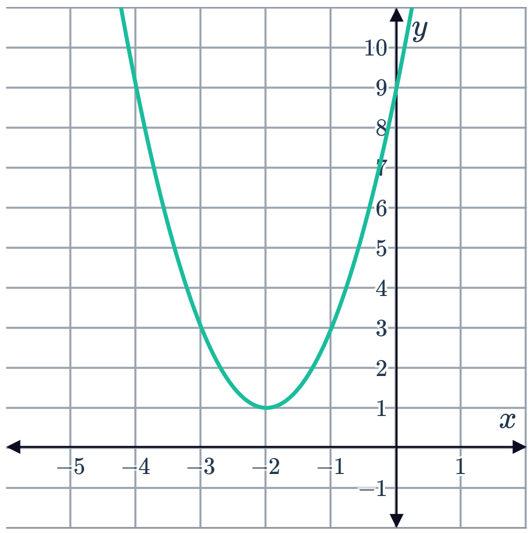
5 Consider the given graph:

- **a** State the x-intercepts.
- **b** State the y-intercept.
- **c** State the maximum value.

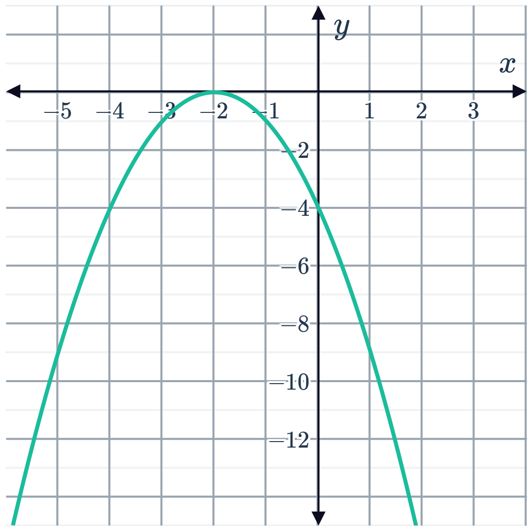


6 Consider the given graph:

- **a** Is the curve concave up or concave down?
- **b** State the y-intercept of the graph.
- **c** What is the minimum value?
- **d** At which value of x does the minimum value occur?
- **e** Determine the interval of x for which the graph is decreasing.



7 Consider the graph of the parabola:



a State the coordinates of the x-intercept.

b State the coordinates of the vertex.

c State whether the following statements are true about the vertex:

i The vertex is the minimum value of the graph.

ii The vertex occurs at the x-intercept.

iii The vertex lies on the axis of symmetry.

iv The vertex is the maximum value of the graph.

8 Suppose that a particular parabola is concave down, and its vertex is located in quadrant 2.

a How many x-intercepts will the parabola have?

b How many y-intercepts will the parabola have?

9 Suppose that a particular parabola has two x-intercepts, and its vertex is located in quadrant 4. Will such a parabola be concave up or concave down?

10 Consider the quadratic function defined in the table on the right:

a What are the coordinates of the vertex?

b What is the minimum value of the function?

x	y
-7	11
-6	6
-5	3
-4	2
-3	3
-2	6
-1	11

11 A vertical parabola has an x-intercept at $(-1, 0)$ and a vertex at $(1, -6)$. Find the other x-intercept.

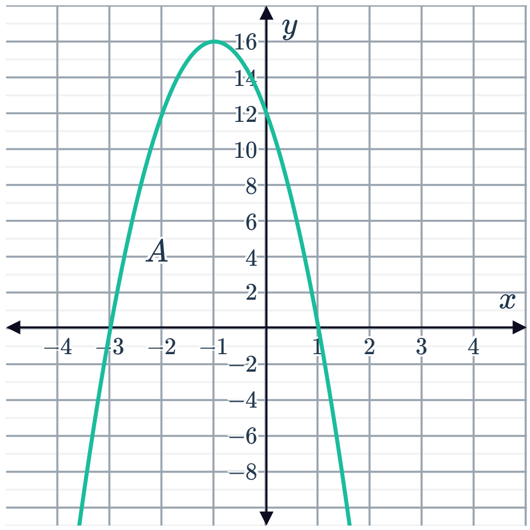
12 State whether the following can be found, without any calculation, from the equation of the form $y = (x - h)^2 + k$ but not from the equation of the form $y = x^2 + bx + c$:

a x-intercepts

b y-intercept

c vertex

13 Quadratic function A is represented graphically as shown. Quadratic function B, which is concave down, shares the same x-intercepts as quadratic function A, but has a y-intercept closer to the origin. Which of the functions has a greater maximum value?



14 State the axis of symmetry of the parabola $y = k(x-7)(x+7)$ for any value of k.

15 Consider the equation $y = 25 - (x+2)^2$. What is the maximum value of y?

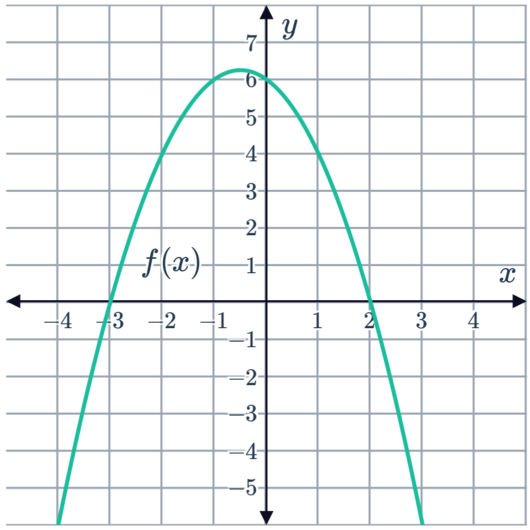
16 Consider the function $y = (14-x)(x-6)$.

- **a** State the zeros of the function.
- **b** Find the axis of symmetry.
- **c** Is the graph of the function concave up or concave down?
- **d** Determine the maximum y-value of the function.

17 Consider the parabola of the form $y = ax^2 + bx + c$, where $a \neq 0$.
Complete the following statement:
The x-coordinate of the vertex of the parabola occurs at $x = \square$. The y-coordinate of the vertex is found by substituting $x = \square$ into the parabola's equation and evaluating the function at this value of x.

18 Find the x-coordinate of the vertex of the parabola represented by $P(x) = px^2 - \frac{1}{2}px - q$.

19 Consider the graph of the function $f(x) = -x^2 - x + 6$:
Using the graph, write down the solutions to the equation $-x^2 - x + 6 = 0$.



20 True or false:

a The quadratic formula can be used to find the y-intercept.

b If the parabola has only one x-intercept , then the x-intercept is also the vertex.

21 Consider the parabola whose equation is $y = 3x^2 + 3x - 7$. Find the x-intercepts of the parabola in exact form.

Graphing quadratics

22 Consider the parabola described by the function $y = -2x^2 + 2$.

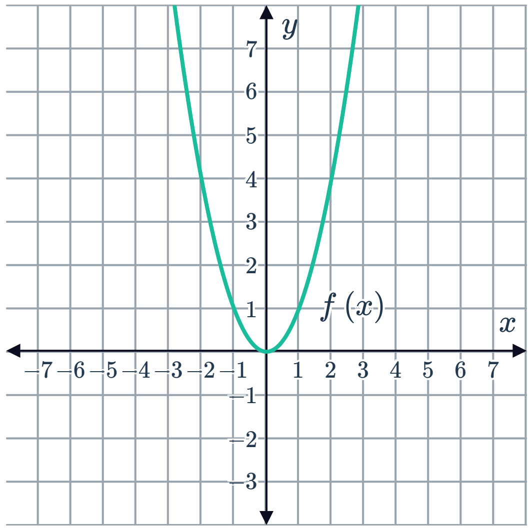
a Is the parabola concave up or down?

b Is the parabola more or less steep than the parabola $y = x^2$?

c What are the coordinates of the vertex of the parabola?

d Sketch the graph of $y = -2x^2 + 2$.

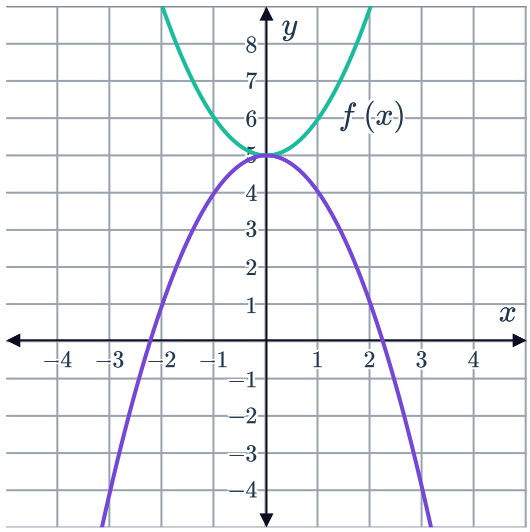
23 A graph of $f(x) = x^2$ is shown on the right.
Sketch the graph after it has undergone transformations resulting in the following functions:



a $g(x) = 4x^2 - 2$

b $h(x) = 2\left(x+4\right)^2$

24 Consider the two graphs. One of them has equation $f(x) = x^2 + 5$.
What is the equation of the other graph?



25 Consider the quadratic function $h(x) = x^2 + 2$.

a Sketch the graph of the parabola $h(x)$.

b Plot the axis of symmetry of the parabola on the same graph.

c What is the vertex of the parabola?

26 Consider the parabola described by the function $y = \dfrac{1}{2}\left(x-3\right)^2$.

a Is the parabola concave up or down?

b Is the parabola more or less steep than the parabola $y = x^2$?

c What are the coordinates of the vertex of the parabola?

d Sketch the graph of $y = \dfrac{1}{2}\left(x-3\right)^2$.

27 Consider the equation $y=(x-3)^2-1$.

- **a** Find the x-intercepts.
- **b** Find the y-intercept.
- **c** Determine the coordinates of the vertex.
- **d** Sketch the graph.

28 Consider the quadratic function $f(x)=-3(x+2)^2-4$.

- **a** What are the coordinates of the vertex of this parabola?
- **b** What is the equation of the axis of symmetry of this parabola?
- **c** What is the y-coordinate of the graph of $f(x)$ at $x=-1$?
- **d** Sketch the graph of the parabola.
- **e** Plot the axis of symmetry of the parabola on the same graph.

29 On a number plane, sketch the shape of a parabola of the form $y=a(x-h)^2+k$ that has the following signs for a, h and k:

- **a** $a>0, h>0, k>0$
- **b** $a<0, h>0, k>0$
- **c** $a>0, h>0, k<0$
- **d** $a<0, h>0, k<0$

30 Consider the parabola $y=(2-x)(x+4)$.

- **a** State the y-intercept.
- **b** State the x-intercepts.
- **c** Complete the table of values:
- **d** Determine the coordinates of the vertex of the parabola.
- **e** Sketch the graph of the parabola.

x	-5	-3	-1	1	3
y					

31 Consider the parabola $y=(x-3)(x-1)$.

- **a** Find the y-intercept.
- **b** Find the x-intercepts.
- **c** State the equation of the axis of symmetry.
- **d** Find the coordinates of the turning point.
- **e** Sketch the graph of the parabola.

32 Consider the parabola $y = x\left(x+6\right)$.

- **a** Find the y-intercept.
- **b** Find the x-intercepts.
- **c** State the equation of the axis of symmetry.
- **d** Find the coordinates of the turning point.
- **e** Sketch the graph of the parabola.

33 Sketch the graph of the following:

- **a** $y = (x+2)(x-3)$
- **b** $y = (x-3)(x+1)$

34 Consider the function $y = \left(x+5\right)\left(x+1\right)$.

- **a** Sketch the graph.
- **b** Sketch the graph of $y = -\left(x+5\right)\left(x+1\right)$ on the same set of axes.

35 Consider the equation $y = x^2 - 6x + 8$.

- **a** Factorise the expression $x^2 - 6x + 8$.
- **b** Hence, or otherwise, find the x-intercepts of the quadratic function $y = x^2 - 6x + 8$
- **c** Find the coordinates of the turning point.
- **d** Sketch the graph of the function.

36 Consider the parabola $y = x^2 + x - 12$.

- **a** Factorise the equation of the parabola.
- **b** Find the x-intercepts of the curve.
- **c** Find the y-intercept of the curve.
- **d** What is the equation of the vertical axis of symmetry for the parabola?
- **e** Find the coordinates of the vertex of the parabola.
- **f** Is the parabola concave up or down?
- **g** Sketch the graph of $y = x^2 + x - 12$.

37 A parabola has the equation $y = x^2 + 4x -$ [redacted]

a Find the y-intercept of the parabola.

b Find the vertex of the parabola.

c Is the parabola concave up or down?

d Hence, sketch the graph of $y = x^2 + 4x - 1$.

38 Consider the quadratic $y = x^2 - 12x + 32$.

a Find the zeros of the quadratic function.

b Find the coordinates of the vertex of the parabola.

c Hence, sketch the graph.

39 Consider the curve $y = x^2 + 6x + 4$.

a Determine the axis of symmetry.

b Hence, determine the minimum value of y.

c Sketch the graph of the function.

40 Consider the function $P(x) = -2x^2 - 8x + 2$.

a Find the coordinates of the vertex.

b Sketch the graph.

41 Consider the equation $y = 6x - x^2$.

a Find the x-intercepts of the quadratic function.

b Find the coordinates of the turning point.

c Sketch the graph.

42 A parabola is described by the function $y = 2x^2 + 9x + 9$.

a Find the x-intercepts of the parabola.

b Find the y-intercept for this curve.

c Find the axis of symmetry.

d Find the y-coordinate of the vertex of the parabola.

e Sketch the graph.

43 Use your calculator to graph the equations below. Then answer the following questions:

i What is the vertex of the graph?

ii What is the y-intercept?

a $y = 4x^2 - 64x + 263$

b $y = -4x^2 - 48x - 140$

44 Use your calculator to graph $y = -3x^2 - 12$.

a What is the vertex of the graph?

b Are there any x-intercepts?

c For what values of x is the parabola decreasing?

45 Using a graphing calculator, sketch the curve $y = x^2 + 6.2x - 7$.

a Determine the axis of symmetry.

b Determine the minimum value of y.

46 Use a graphing calculator to sketch the parabola $y = -2x^2 + 16x - 24$.

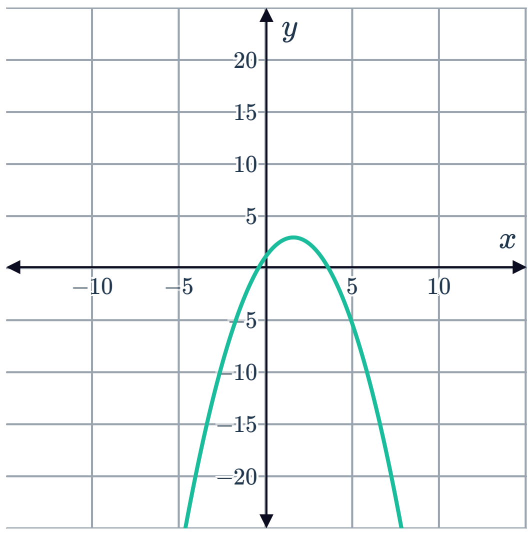
a Find the x-intercepts of the parabola.

b Find the y-intercept of the parabola.

c Find the axis of symmetry of the parabola.

d Find the y-coordinate of the vertex of the parabola.

47 Consider the function $y = -0.72x^2 + \sqrt{5}x + 1.21$.



a Find the x-coordinate of the vertex to two decimal places.

b Find the y-coordinate of the vertex to two decimal places.

c Is the graph shown a possible viewing window on your calculator that shows the vertex and the x-intercepts?

d Use a graphing calculator to find the x-intercepts to two decimal places.

48 Consider the function $y = 0.91x^2 - 5x - \sqrt{\ }$ [illegible]

a Find the x-coordinate of the vertex to two decimal places.

b Find the y-coordinate of the vertex to two decimal places.

c Is the graph shown a possible viewing window on your calculator that shows the vertex and the x-intercepts?

d Use a graphing calculator to find the x-intercepts to two decimal places.